NYU 上海 SHANGHAI 纽约大学

Introduction to Computer Programming
Fall 2026
Mock Midterm

First Name	
Last Name	
NetID	

Instructions

- **Duration**: 90 minutes
- **Max score**: 100 pts

- Computers, calculators, phones, textbooks or notebooks are **not allowed**
- When writing programs, make very clear your line **indentations** (there are some indentation levels in the areas where you will write code). If indentation is not perfectly clear, the grader will consider "no indent".
- We can **not answer** questions during the test.
- If you are unsure of a problem, please write down your assumptions, and proceed with it. We will grade accordingly.
- **Do not** use lists, string methods (such as .split() or .count()), dictionary, set or define functions. Use only concepts covered in class.

Grading

Questions	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**
Points	$8'$	$8'$	$24'$	$10'$	$10'$	$10'$	$15'$	$15'$
Score								
Total								

Question 1 | Boolean Expression

Evaluate the following boolean expression:

`not True or True`	`True`
`3 != '3' or 3 > 5`	`True`
`1 < ('Bee' == 'bee') and 5//2  < 1`	`False`
`not False or False and True`	`True`

Question 2 | Binary and Hexadecimal Representaion

Complete the table below by converting each given number into the other two number systems: decimal, binary, and hexadecimal.

Decimal	Binary	Hexadecimal
45	`0b101101`	`0x2d`
`63`	`0b111111`	3F

Question 3 | Predict the Output

Program 1

```
print(23 // 4)
print(23 % 4)
```

Output 1

```
5
3
```

Program 2

```
a = 5
b = 2

print(a, b, sep=" + ", end=" = ")
print(a + b)
print(f"{a / b:.2f}")
```

Output 2

```
5 + 2 = 7
2.50
```

Program 3

```
word = "PROGRAM"

print(word[1:5])
print(word[-2:2])
print(word[-3:])
print(word[::-2])
print(word[-1:3:-1])
```

Output 3

```
ROGR

RAM
MROP
MAR
```

Program 4	Output 4

`for i in range(1, 8):`
`  if i % 3 == 0:`
`  continue`
`  if i > 5:`
`  break`
`  print(i)`
 	
`print("End")`
	1
	2
	4
	5
	End

Program 5	**Output 5**

`n = 19`
 	
`while n > 0:`
`  print(n)`
`  n //= 2`
	19
	9
	4
	2
	1

Program 6	**Output 6**

`result = ""`
 	
`for ch in "a1b2c3":`
`  if ch in "123":`
`  result = ch + result`
 	
`print(result)`
	321

Program 7	**Output 7**

`x = 12`
 	
`if x % 3 == 0:`
`  print("A")`
`  if x % 5 == 0:`
`  print("B")`
`  else:`
`  print("C")`
`else:`
`  print("D")`
	A
	C

Program 8	**Output 8**

`total = 0`
 	
`for i in range(2, 8, 2):`
`  total += i`
`  print(i, total)`
	2 2
	4 6
	6 12

Question 4 | Predict the Output

The string seats represents a row of cinema seats. X indicates an occupied seat, and `.` indicates an available seat. Write the exact output of the following program.

```
seats = "X..X...X...."

for group_size in range(2, 6):
    position = 0
    available = 0

    while position < len(seats):
        if seats[position] == ".":
            available += 1
        else:
            available = 0

        if available == group_size:
            first_seat = position - group_size + 2
            print(f"Group {group_size}: seats "+
                  f"{first_seat}-{position + 1}")
            break

        position += 1

    if available < group_size:
        print(f"Group {group_size}: no suitable seats")
```

Your answer :

```
Group 2: seats 2-3
Group 3: seats 5-7
Group 4: seats 9-12
Group 5: no suitable seats
```

Question 5 | Triangle Classifier

Write a Python program that reads three positive integers representing the side lengths of a triangle.

Classify the input as follows:

- **Invalid triangle:** The sum of any two sides is less than or equal to the third side.
- **Equilateral triangle:** All three sides are equal.
- **Isosceles triangle:** Exactly two sides are equal.
- **Scalene triangle:** All three sides are different.

Example Output (user input is in bold)**:**

Example 1:

Enter side 1: ***5***
Enter side 2: ***5***
Enter side 3: ***3***

Isosceles triangle

Example 2:

Enter side 1: ***2***
Enter side 2: ***3***
Enter side 3: ***5***

Invalid triangle

Your code:

```
a = int(input("Enter side 1: "))
b = int(input("Enter side 2: "))
c = int(input("Enter side 3: "))

if a + b <= c or a + c <= b or b + c <= a:
    print("Invalid triangle")
elif a == b == c:
    print("Equilateral triangle")
elif a == b or a == c or b == c:
    print("Isosceles triangle")
else:
    print("Scalene triangle")
```

Question 6 | Digit Counter

Write a Python program that reads a positive integer `n` and a threshold `k`, then displays how many digits of `n` are **greater than or equal to** `k`.

Example Output (user input is in bold)**:**

Enter a positive integer: ***431663334231***
Enter a threshold (0-9): ***3***
Count: 9

Your code:

```
n = int(input("Enter a positive integer: "))
k = int(input("Enter a threshold (0-9): "))

count = 0

while n > 0:
    digit = n % 10

    if digit >= k:
        count += 1

    n //= 10

print("Count:", count)
```

Question 7 | Message Compressor

A simple message compressor replaces each consecutive group of identical characters with the character followed by the number of times it appears. Write a program that reads a message and prints its compressed form.

Example Output (user input is in bold)**:**

Example 1:

Enter a message: ***aaabbcaaaa***
Compressed: a3b2c1a4

Example 2:

Enter a message: ***AAAaa***
Compressed: A3a2

Your code:

```
message = input("Enter a message: ")
compressed = ""
i = 0

while i < len(message):
    character = message[i]
    count = 1
    i += 1

    while i < len(message) and message[i] == character:
        count += 1
        i += 1

    compressed += character + str(count)

print("Compressed:", compressed)
```

Question 8 | Dice Duel

Two players, Alice and Bob, play a dice game. In each round, the computer rolls one six-sided die for each player.

- If Alice's roll is higher, Alice earns one point.
- If Bob's roll is higher, Bob earns one point.
- If the rolls are equal, neither player earns a point. Print `Tie! Roll again.` and roll again for both players in the same round.
- The first player to earn **3** points wins.

Write a Python program that simulates the game and follows the output format shown in the example below.

Example Output:

```
Round 1
Alice rolls 4, Bob rolls 4
Tie! Roll again.
Alice rolls 6, Bob rolls 2
Score: Alice 1, Bob 0

Round 2
Alice rolls 1, Bob rolls 5
Score: Alice 1, Bob 1

Round 3
Alice rolls 3, Bob rolls 2
Score: Alice 2, Bob 1

Round 4
Alice rolls 2, Bob rolls 2
Tie! Roll again.
Alice rolls 5, Bob rolls 3
Score: Alice 3, Bob 1

Alice wins!
Rounds played: 4
```

Your code:

```
import random

alice_score = 0
bob_score = 0
rounds = 0

while alice_score < 3 and bob_score < 3:
    rounds += 1
    print(f"Round {rounds}")

    alice_roll = random.randint(1, 6)
    bob_roll = random.randint(1, 6)
    print(f"Alice rolls {alice_roll}, Bob rolls {bob_roll}")

    while alice_roll == bob_roll:
        print("Tie! Roll again.")

        alice_roll = random.randint(1, 6)
        bob_roll = random.randint(1, 6)
        print(f"Alice rolls {alice_roll}",
              f"Bob rolls {bob_roll}")

    if alice_roll > bob_roll:
        alice_score += 1
    else:
        bob_score += 1

    print(f"Score: Alice {alice_score}, Bob {bob_score}")

if alice_score == 3:
    print("Alice wins!")
else:
    print("Bob wins!")

print(f"Rounds played: {rounds}")
```